



UNIVERSITY OF GOTHENBURG
SCHOOL OF BUSINESS, ECONOMICS AND LAW

Individual assignment:
Using sentiment data to predict future
stock market performance

Oliver Nilsson
gusnilolaa@student.gu.se

Course: GM1413 HT24 Advanced Data Analysis

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1 Introduction

This study analyzes the impact of sentiment scores from companies' earnings conference calls on stock market performance.

2 Data Sources

The analysis data is based on a given list of randomly generated tickers from the course assignment (see Appendix 1).

2.1 Transcript Data

Transcript data, provided as course material, includes key information such as date, ticker, participants, description, and speech content.

2.2 Loughran-McDonald Sentiment Dictionary

The Loughran-McDonald sentiment dictionary is used to define if a word in the transcript is positive, negative or unclear.

2.3 WRDS Data

Data on trading volume and returns for all tickers and the S&P 500 have been downloaded from the WRDS database, specifically from Compustat – Capital IQ and CRSP.

2.4 Yahoo Finance

Each ticker's sector is fetched from Yahoo Finance using the Python library `yfinance`.

3 Method

Note: Ticker 'FB' has been changed to 'META' on June 9, 2022, following the company's rebranding from Facebook to META. This change has been applied in the code and to the transcript pickle file name to further ensure correct data fetching (Meta 2022).

3.1 Extracting Text and Calculating Sentiment Scores

The transcripts for each company's earnings call are extracted by iterating through the transcript data. Each transcript has a JSON data structure containing components such as speech, timestamps, participant details etc.

To process the transcripts, the NLP-model spaCy is used for lemmatization, text preprocessing and tokenization. The data is processed in batches to reduce memory usage and improve script efficiency. The Loughran-McDonald sentiment dictionary is applied to classify words in the transcript as positive, negative, or uncertain, with sentiment scores calculated for each transcript.

3.2 Role Defining using NLP Llama 3.1

After pre-processing the data, it was found that the transcripts data defining participants role were very unstructured and had no pattern. This makes it difficult to manually check for key words in the role description and get a significant role count. These conditions create a good use case for use of Natural Language Processing (NLP).

In the Role Defining the description-data is prompted to the NLP to give the participant a role of: Manager, Analyst or Other. All results from the NLP are stored in a JSON-file with the description as key and role as value. When the descriptions are being processed, it is first checking if that description already has been prompted and therefore stored in the JSON-file. This streamlines the process and prevents redundancy where same description is being prompted multiple times.

3.3 Data Processing

Numerical data from WRDS and Yahoo Finance were cleaned, merged, and formatted for to match analysis requirements.

3.4 Cumulative Abnormal Return using Event Window

Cumulative Abnormal Return (CAR) is calculated by comparing the stock return with the market return, which in this case being the S&P 500 returns. The event window is defined as two trading days before to two trading days after the earnings call is defined, allowing for the capture of the immediate market reaction.

Expected return for each day in the event window is calculated using the market model which assumes that the stock returns to be in line with the market returns. A linear regression is run on the historical estimation of the 30 trading days before to estimate the relationship between the stock's returns and the S&P 500 returns.

$$R_{i,t} = \alpha + \beta R_{S\&P\ 500,t} + \epsilon_{i,t}$$

where:

- $R_{i,t}$: return of the stock i on day t
- $R_{S\&P\ 500,t}$: return of S&P 500 on day t
- α and β : regression coefficients that estimates the stock's sensitivity to the market

$$AR_{i,t} = R_{i,t} - \beta R_{S\&P\ 500,t}$$

$$CAR_i = \sum AR_{i,t}$$

The result, CAR_i , is a measure of the Cumulative Abnormal Return for the Event Window, which represents the effect of the earnings call.

3.5 Panel OLS-regression

To analyze the impact of net sentiment on CAR, Panel OLS-regression model was used. The model allows us to control for both entity (company) and time effects, providing a more robust understanding of how sentiment affects CAR.

The combined model was run across all sectors to understand the general relationship between sentiment and CAR.

Sector-specific models were also run for those sector that had more than one ticker to be able to run Panel OLS-regression across multiple companies and dates. This was made to discover how the impact of sentiment varies by industry.

3.6 Visualization for Specific Sectors

The results are visualized using bar plots to present R-squared values and p-values for each sector. The p-values were transformed using negative log10 to compress very small values, making it easier to visually see differences in statistical significance across sectors. This visualization helps to highlight which sectors are most significantly impacted by net sentiment.

4 Model Specification

4.1 Dependent and Independent Variable

The dependent variable is set to Cumulative Abnormal Return (**CAR**) and is used as a proxy to measure the equity capital market reaction to earnings conference calls.

The independent variable is **net_sentiment**, which measures the net sentiment from earnings conference calls.

4.2 Control Variables

- **manager_ratio**: Represents proportion of managers participating in the earnings call.
- **analyst_ratio**: Represents proportion of analysts participating in the earnings call.
- **other_ratio**: Represents proportion of other participants in the earnings call.
- **volume**: Represents the trading volume of the stock, capturing market activity levels.
- **stock_volatility**: Represents the volatility of the stock, which could influence abnormal returns.
- **lag_stock_return**: Previous period's stock return, used to control for momentum or persistence in returns.
- **lag_sp500_return**: The lagged return of the S&P 500, used to account for market-wide movements that might influence individual stocks.

4.3 Regression Model Choice

The Panel OLS-regression model was selected due to the panel nature of the dataset, which contains repeated observations for multiple companies over multiple time periods. This allows to control for both entity effects (differences across firms) and time effects (changes over time). The data variable is set to include only year to better fit the analysis and data.

4.4 Variable Definitions and Expected Relationships

- **Net Sentiment (net_sentiment)**: Expected to have a positive relationship with CAR.
- **Manager Ratio (manager_ratio), Analyst Ratio (analyst_ratio) and Other Ratio (other_ratio)**: These ratios capture sentiment from different sources in the earnings call. While a positive relationship is generally expected, negative coefficients in some cases could indicate market skepticism towards overly optimistic communication.

- **Volume (volume):** Expected to be related to CAR since higher trading volumes may indicate stronger investor reactions. The direction could vary.
- **Lagged Stock Return (lag_stock_return):** Expected to show **momentum** effects, potentially leading to positive or negative impacts depending on recent performance trends.
- **Lagged S&P 500 Return (lag_sp500_return):** Used as a control for overall market conditions. It is expected that positive market-wide movements could lead to lower abnormal returns as stock price increases are already accounted for.

5 Results and Analysis

5.1 Combined Sector Analysis

The combined Panel OLS-regression model shows the relationship between net sentiment and Cumulative Abnormal Return (CAR) across all sectors (see Appendix 2). The R-squared value is 0.0748, indicating that the model explains 7.48% of the variance in CAR. This low R-squared is typical for financial models involving abnormal returns due to high unpredictability. With respect to how many factors that affects stock performance it might not be reasonable to be able to create a model with high R-squared value that is include net sentiment as a significant variable.

The R-squared (Between) value is 0.4498, suggesting the model captures 44.98% of the variation between companies, while the R-squared (Within) value is 0.0708, indicating limited effectiveness in explaining time-based variation within firms.

Net sentiment has a positive and significant effect on CAR (p-value = 0.0059), supporting the hypothesis that positive sentiment from earnings calls slightly increases abnormal returns.

The analysis shows that net sentiment significantly impacts CAR, supporting the assignment's goal of understanding how sentiment influences market reactions. The significance of control variables highlights the complexity of factors affecting CAR.

5.2 Sector-Specific Analysis

The sector-specific analysis highlights differences in how net sentiment and other factors impact CAR across industries (see Appendix 3). R-squared values and significance levels vary between sectors, indicating different model effectiveness and sentiment roles.

- **Communication Services:** Shows a relatively low R-squared and that net_sentiment is not significant.
- **Financial Services:** Shows a relatively high R-squared and significant net sentiment, indicating sentiment plays a role in influencing CAR. This may be because financial markets could be more sensitive to investor sentiment and forward-looking statements, which can directly impact valuation.
- **Industrials:** The R-squared value was relatively high, but net sentiment was not significant, suggesting that other factors may play a larger role in determining CAR for industrial companies.
- **Healthcare and Technology:** Show lower R-squared values and no significant net sentiment, suggesting other factors play a larger role, such as regulatory news or innovation.
- **Consumer Sectors (Cyclical and Defensive):** Net sentiment impact varies. Consumer Cyclical showing that net_sentiment is statistically significant. Consumer Defensive showing that net_sentiment is not significant, possible due to differing consumer behavior.

The analysis shows that net sentiment effects on CAR are clearly differing depending on sector. Sectors like Financial Services and Industrial are more influenced by sentiment, while Healthcare and Technology are affected by other factors. These results show the need to consider sector characteristics when analyzing sentiment's impact on stock performance.

6 References

Loughran, M. and McDonald, B., 2018. *LoughranMcDonald_SentimentWordLists_2018.xlsx*. [course material]

META (2022). *Meta Platforms, Inc. to Change Ticker Symbol to 'META' on June 9*. <https://investor.fb.com/investor-news/press-release-details/2022/Meta-Platforms-Inc.-to-Change-Ticker-Symbol-to-META-on-June-9/default.aspx> [2022-05-31]

Wharton Research Data Services, (2024). *Compustat – Capital IQ and CRSP Data*. [course material]

Yahoo Finance (2024). Data retrieved using Python yfinance library. <https://finance.yahoo.com/> [2024-11-11]

7 Appendices

7.1 Tickers

['BX', 'CRM', 'NKE', 'CHTR', 'SE', 'CVS', 'QCOM', 'WFC', 'DIS', 'TSLA', 'GOOG', 'MA', 'PEP', 'VZ', 'TMUS', 'UNP', 'MS', 'NVS', 'NVO', 'MCO', 'JNJ', 'FB', 'MCD', 'GM', 'AMT', 'MRK', 'CVX', 'AMZN', 'FDX', 'AAPL', 'ISRG', 'NVDA', 'KO', 'LMT', 'SBUX', 'BA', 'HD', 'ADBE', 'MET', 'BMY', 'TD', 'LLY', 'JPM', 'AXP', 'DUK', 'TGT', 'EL', 'TXN', 'F', 'AMGN', 'INTC', 'ASML', 'ORCL', 'HON', 'SNY', 'C', 'WMT', 'CMCSA', 'EBAY', 'TMO']

7.2 Combined Panel OLS Results

PanelOLS Estimation Summary

Dep. Variable:

CAR

R-squared:

0.0748

Estimator:

PanelOLS

R-squared (Between):

0.4498

No. Observations:

4360

R-squared (Within):

0.0708

Date:

Mon, Nov 11 2024

R-squared (Overall):

0.0775

Time:

19:56:35

Log-likelihood

6677.0

Cov. Estimator:

Clustered

F-statistic:

49.366

Entities:

60

P-value

0.0000

Avg Obs:

72.667

Distribution:

F(7,4277)

Min Obs:

33.000

F-statistic (robust):

25.487

Max Obs:

142.00

P-value

0.0000

Time periods:

17

Distribution:

F(7,4277)

Avg Obs:

256.47

Min Obs:

3.0000

Max Obs:

455.00

Parameter Estimates

	Parameter	Std. Err.	T-stat	P-value	Lower CI	Upper CI
net_sentiment	0.1672	0.0607	2.7568	0.0059	0.0483	0.2862
manager_ratio	-0.0165	0.0033	-4.9901	0.0000	-0.0231	-0.0100
analyst_ratio	-0.0173	0.0045	-3.8207	0.0001	-0.0261	-0.0084
other_ratio	-0.0106	0.0052	-2.0636	0.0391	-0.0207	-0.0005
volume	-8.119e-11	2.149e-11	-3.7771	0.0002	-1.233e-10	-3.905e-11
stock_volatility	0.8284	0.1904	4.3504	0.0000	0.4551	1.2018
lag_stock_return	0.6218	0.0721	8.6232	0.0000	0.4804	0.7632
lag_sp500_return	-0.7628	0.1019	-7.4837	0.0000	-0.9626	-0.5630

F-test for Poolability: 1.2557

P-value: 0.0686

Distribution: F(75,4277)

Included effects: Entity, Time

7.3 Sector-Specific Panel OLS Visualizations

